



POPS User Manual

Rev 8



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Laser Safety Warnings

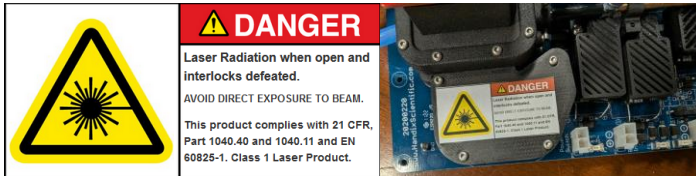
All variants of the POPS are Class 1 Laser Products with an embedded Class 3B 405 nm, 70 mW laser. Under normal operation there is no exposure to laser radiation.

Certification Label(s) on Instrument: Location(s): Top of instrument on Plastic main face, and on top of sealed optical chamber.

Location 1:



Location 2:



Identification Label on Instrument: Location on top of Instrument

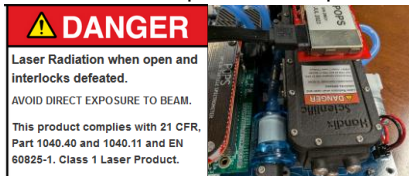


Label(s) for defeatably interlocked protective housings:

Location 1: On top of Sealed optical chamber



Location 2: On top of Sealed optical chamber



WARNING: The use of controls, adjustments or procedures other than those specified in this manual may result in exposure to hazardous laser radiation.

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Introduction

The Portable Optical Particle Spectrometer (POPS) is a light weight, high performance optical particle counter. It counts and measures the optical size of sampled particles using single-particle light scattering between approximately 130 nm and 3 μm . The POPS comes in several packages depending on the intended application. The POPS-1100 model does not come with a standard enclosure, has various options depending on application requirements, and is intended to be a flexible measurement package that can be installed in a variety of sensor platforms. The POPS-1110 model includes an aluminum-coated foam enclosure and hardware for autonomous airborne operation intended for use with unmanned balloon systems. The POPS-1120 model includes a hardened weatherproof enclosure and other hardware for outdoor and bench top applications.

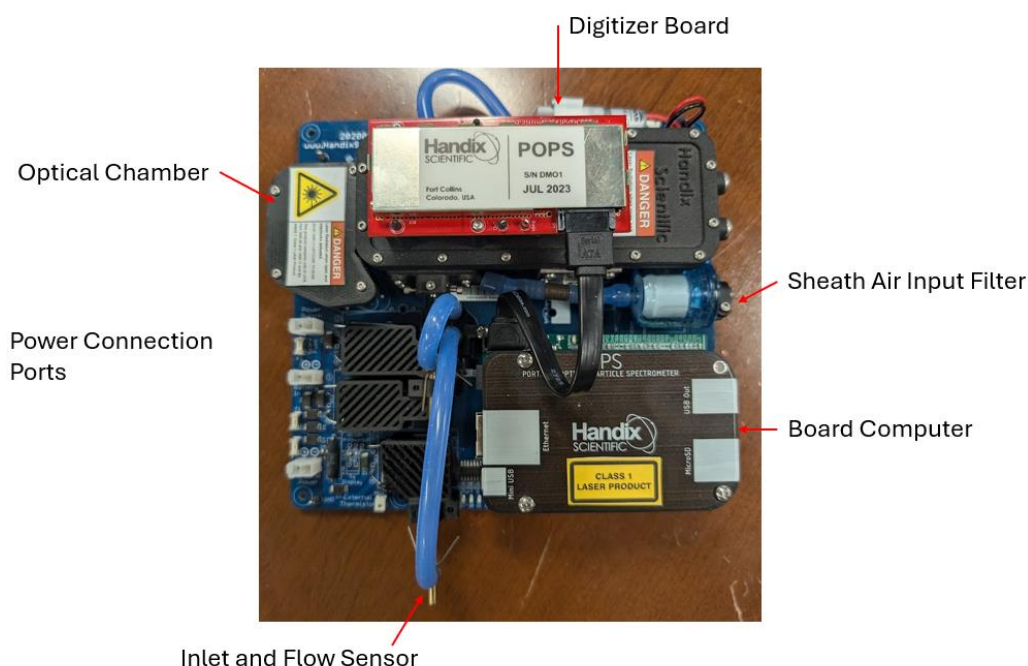


Figure 1. Overview of the core POPS components, with data system and optical chamber mounted to baseplate.

The POPS was originally developed by the Chemical Sciences Division of the US National Oceanic and Atmospheric Administration (NOAA) and is licensed to Handix Scientific Inc.

POPS optical particle detection chamber

The optical chamber consists of the main sampling enclosure, which houses the laser diode, focusing optics, light collection mirror, laser beam dump, photo detector and inlet and exhaust ports. The laser beam produced by an ~ 70 mW 405 nm laser diode is aligned to the horizontal center of the optical chamber, shown in Figure 2. The vertical position of the laser beam is adjusted so that the center of the profile is aligned with the inlet nozzle tip. The laser alignment is adjusted using three screws located on the rear of the laser diode assembly and comes pre-aligned.

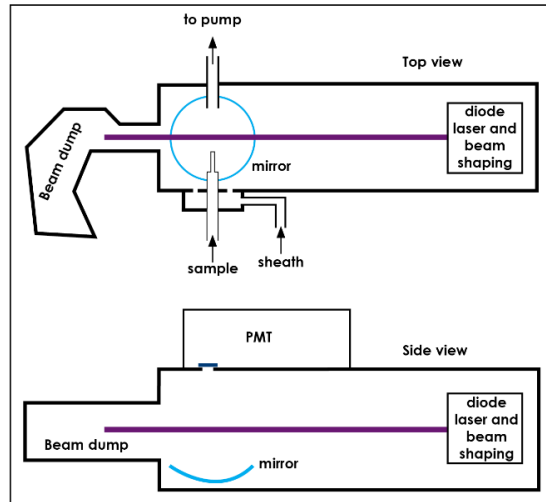


Figure 2. Schematic showing top and side views of the POPS optical particle detection chamber and main components.

The inlet is a 1/8" OD (0.069" ID) stainless steel tube containing a 0.4" long nozzle (1/16" OD, 0.032" ID) that is inserted 0.1" into the main tube. The nozzle tip is positioned within 0.5 cm of the laser beam. The exhaust is positioned approximately 1 cm from the exhaust beam. The sample jet/detection region is focused onto the sensing area of a photomultiplier tube located at the top of the optical chamber by a spherical mirror.

A series of four pairs of slits prevents stray light from entering the detection area and introducing background light. Each pair of slits is adjusted using two setscrews that are accessed from the top and side of the optical chamber.

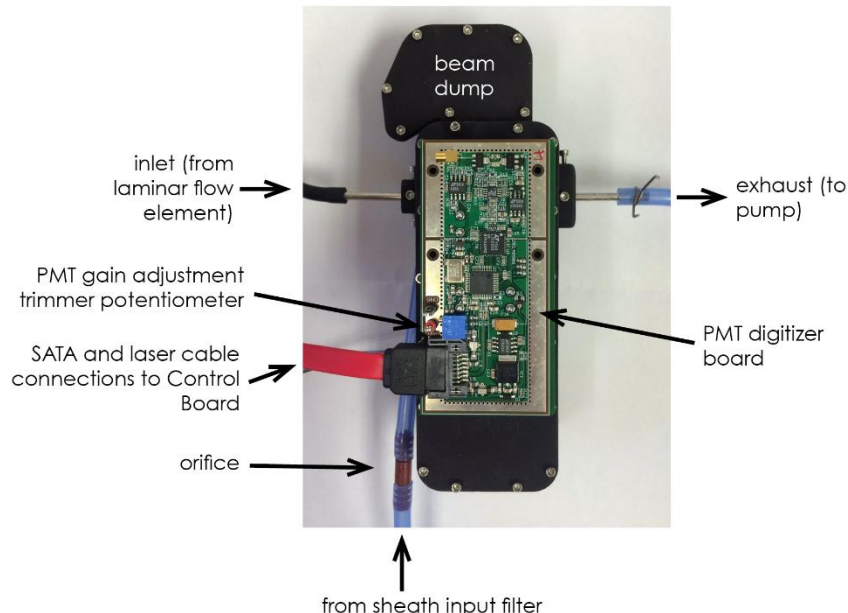


Figure 3. Top view of the POPS optical chamber. Note: Other versions of the chamber have an integrated outlet port rather than the stainless steel tubing shown in the picture.

The circuit board mounted to the top of the optical chamber is the signal processing and digitizer for the raw light scattering signals measured by the photo detector, shown in Figure 3. It is connected to the instrument support hardware through a SATA cable. It also has a trimmer potentiometer (trimpot) for adjusting the gain on the photomultiplier tube signal amplification circuit. The gain is set at the factory to produce a raw signal of ~3160 analog-to-digital counts for a 508 nm polystyrene latex sphere calibration particle, which has been found to produce the best signal-to-noise over the detectable size range (Gao et al., 2016).

Data acquisition and control package

The data acquisition and control package consist of a Beaglebone Black Industrial single-board computer running Linux (referred to in this manual as the POPS Computer), custom POPS support board (referred to as the POPS Cape), and custom POPS baseplate board (referred to as the POPS Baseplate). More details on the Beaglebone Black (BBB) device may be found at <https://beagleboard.org/black>. The POPS has an “Industrial” version of the board rated for temperatures between -40 to +85 C. A 32 GB micro-SD memory card is supplied and pre-installed with the POPS Computer for data storage.

The POPS Cape, shown in Figure 4 is an interface between the individual component systems (detector board, laser, pump, flow sensor, user controls) and the POPS Computer. It has many functions, including providing a current source for the POPS laser diode, passing digitized raw particle signals from the photo detector board to the computer and controlling the pump voltage and sample flow.

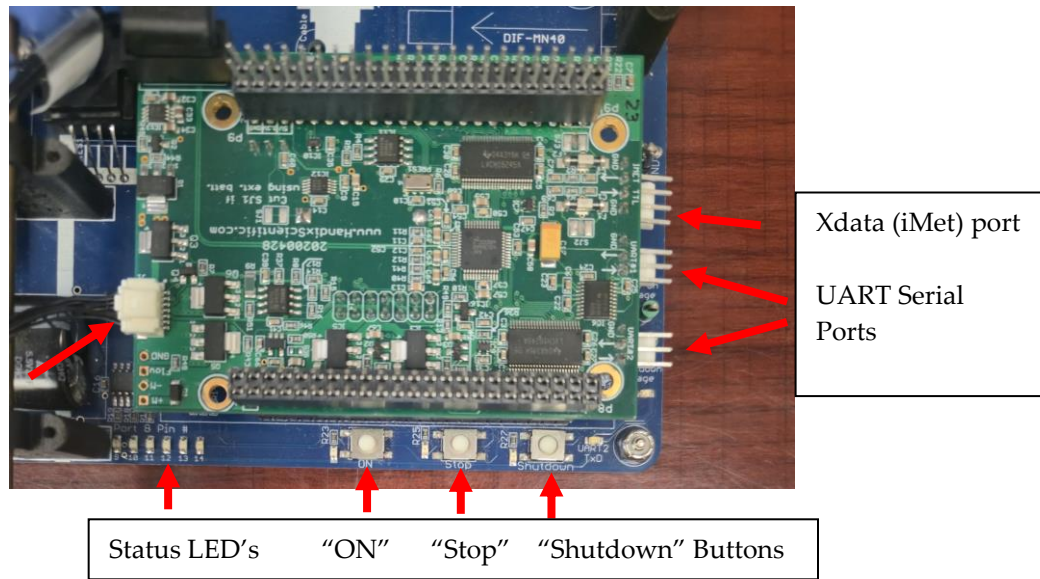


Figure 4. Top view of the POPS Cape and important baseplate components, highlighting the location of the POPS Cape connections, and baseplate mounted Status LEDs and buttons for operations.

The baseplate is the main chassis for the other instrument components and has several input and output features for basic operation and troubleshooting. The baseplate has a series of LEDs located along the edge closest to the inlet and laminar flow element (shown in Figure 4) that illuminate when the system has been powered on (LEDs labeled 9 and 10), the software has been loaded (LED 11) and the system is operating normally (LED 13 will blink every second). Any faults, typically associated with an over-voltage on one the analog input signals, will cause LED 14 to illuminate, which is red. This error will not necessarily prevent the system from acquiring data but persistent problems, such as a faulty sample pump or large leak, will cause the repeated illumination of the error LED and users should contact Handix Scientific for assistance.

The baseplate has connections for ground-based DC power, battery DC power, and a power switch which is used in conjunction with the battery. Instruments intended for balloon and outdoor applications (POPS-1110 and 1120) are powered using battery power and come installed with a switch to toggle power to the POPS. All other applications should use the normal “ground power” connection, which will power on the instrument as soon as DC power is supplied (usually from an AC to DC power plug).

Turning off the instrument consists of disconnecting the power or pushing the power switch depending on the model of POPS. Another method to shutdown the instrument is using the “STOP” and “SHUTDOWN” buttons on the baseplate. In current versions of the instrument there is no difference in functionality between the “STOP” and “SHUTDOWN” buttons. Each sends an interrupt signal to the acquisition program and triggers a shutdown of the POPS computer. The shutdown operation can take several seconds. If the system fails to shut down, try pressing the button again or pressing and holding the button until the system shuts down. When the system has shutdown safely the LEDs will no longer be illuminated.

Vacuum pump, sample and exhaust tubing

The complete POPS flow schematic is shown in Figure 5. The POPS uses a miniature rotary vane pump to produce a small pressure drop in the optical chamber and pull sample and sheath flow into the cavity. The pump speed is controlled through a feedback loop running and is automatically adjusted to maintain a near-constant sample flow rate through the laminar flow element connected to the POPS inlet. The sheath flow

is proportional to the pressure drop in the optical chamber and is passively controlled through an orifice. The sheath air is taken from the ambient air, drawn through a filter and orifice before passing through six holes located around the sample inlet nozzle.

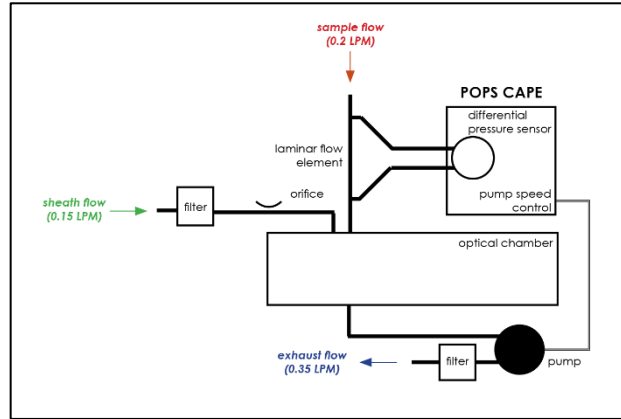
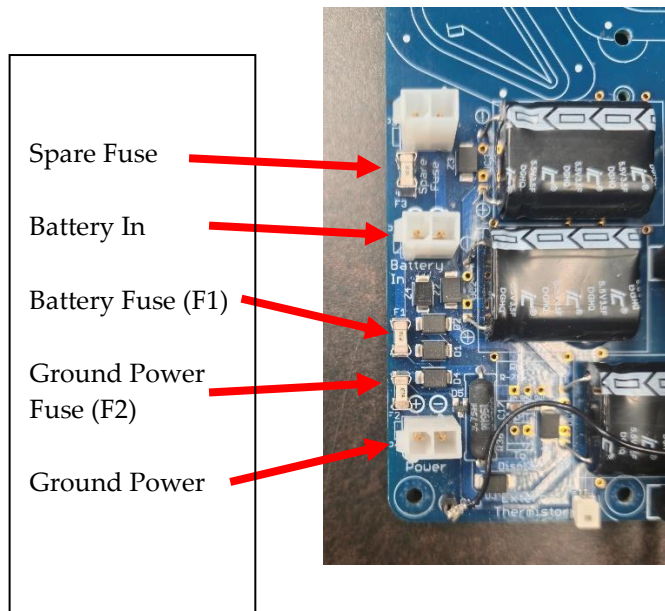


Figure 5. POPS sample, sheath and exhaust flow schematic. The filter on the exhaust pump is optional.

Power

The POPS requires 11-15 VDC input power, supplied either by a DC power supply (e.g., a wall block AC-DC power supply) or an external battery module. Each method of input power has a 2A fuse to protect the POPS from excess power mounted on the POPS baseplate which are labeled F1 for battery power and F2 for ground power. Although rare, it is possible for a fuse to blow, preventing the POPS from being turned on when supplied with the necessary power. These fuses can be replaced with the spare that is located on the baseplate as well, marked as “F3”, using small pliers to remove the old fuse and replace it with the new one. Contact Handix Scientific for further instruction.



Unpacking and Setup

Carefully inspect the shipping container for any signs of damage and verify all major components of the instrument package are included and match the description provided on the packing slip. POPS-1110 and POPS-1120 versions of the instrument will come pre-assembled, and the POPS-1100 version may include separate components.

POPS Component Overview

Major components include:

- POPS optical particle detection chamber
- Data acquisition and instrument control hardware (POPS computer, POPS Cape)
- Vacuum pump, sample and exhaust tubing
- AC/DC universal input power supply and/or battery packs (model dependent)
- RJ45 ethernet cable (model dependent)
- Mini USB cable
- microSD card
- USB Flash Drive

Operating the POPS

Base configuration (POPS-1100):

To begin sampling, first verify the POPS inlet is not obstructed and the micro-SD card is fully inserted into the POPS Computer, then plug in the instrument. The LEDs on the Baseplate should automatically illuminate after a few seconds indicating the POPS startup sequence. The pump should begin operating within a minute and quickly settle on a steady operating speed to maintain a constant sample flow rate. After another minute the POPS data acquisition software will automatically load the current configuration, begin acquiring and saving particle data to file and send housekeeping and binned data via UDP and various serial protocols to any connected external systems. To turn off the system, you may either use the “STOP” or “SHUTDOWN” buttons labeled on the baseplate, or simply disconnect the power. If using the buttons, you may need to press and hold until the system status LEDs turn off and the pump turns off as well, and then power may be removed.

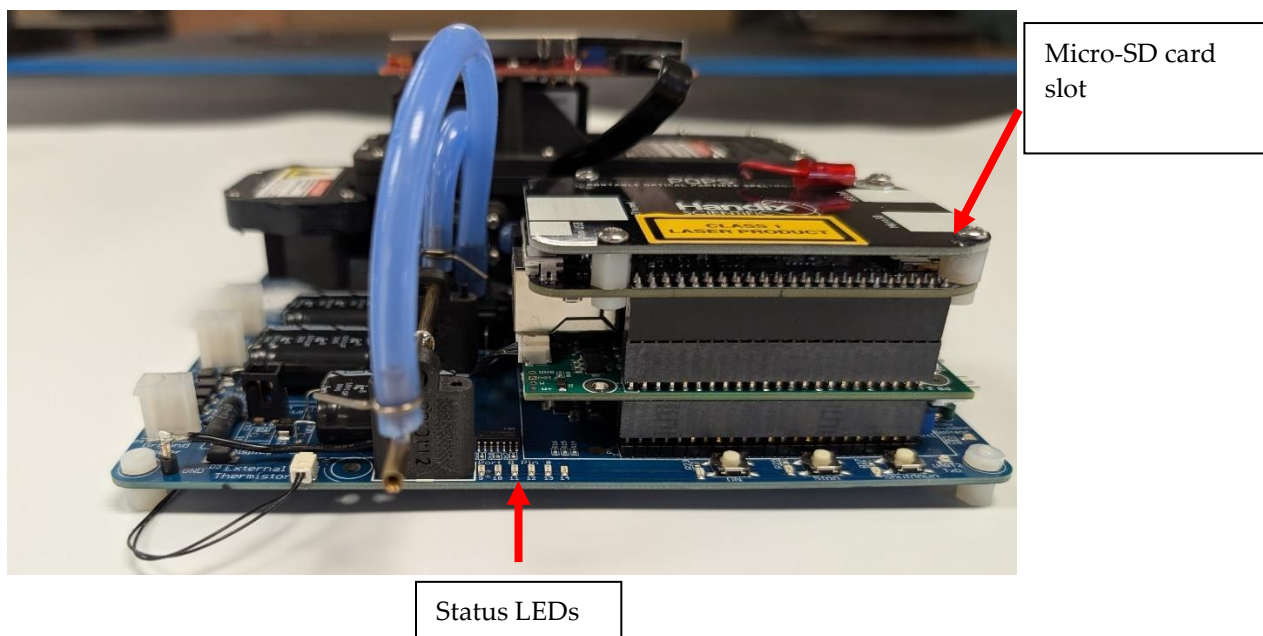


Figure 7. Locations of the micro SD card (inserted into slot on POPS Computer), and system status LEDs.

Balloon version (POPS-1110):

Setup: Balloon-configured instruments are shipped in a foam enclosure with a pass-through for the sample inlet tubing. The instrument is supplied with a short brass 1/8" outer diameter tube that should be inserted firmly into the inlet housing and connected via the supplied static tubing. It is easiest to do this by opening the foam box and pushing the inlet against the housing while holding the instrument assembly with your other hand. Connect to the battery to the external battery connector port on the power switch, as shown in Figure 8. The power switch button can easily be pressed from the exterior of the box to turn the POPS on and off.

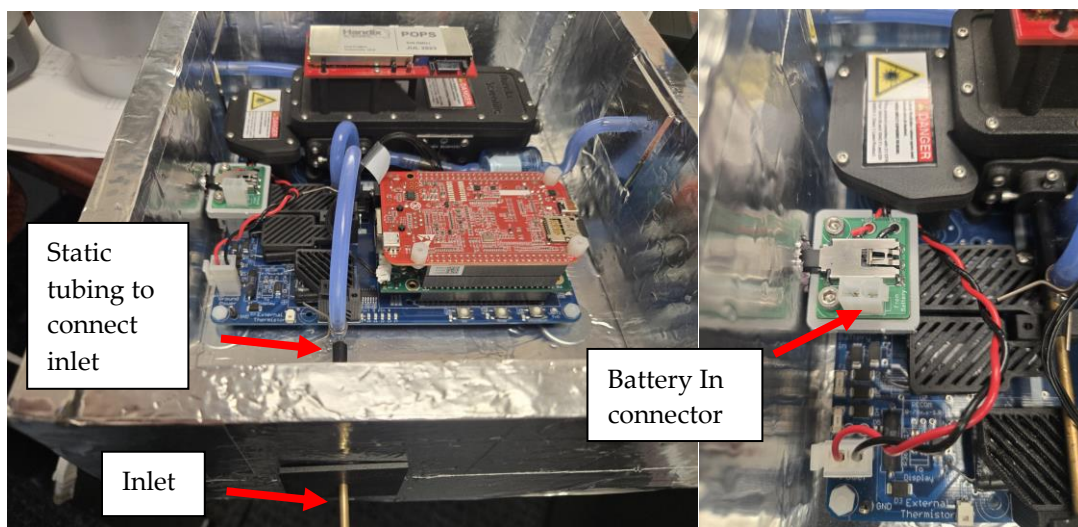


Figure 8. Balloon configuration (left) and Power switch connector location (right).

Verify that the micro-SD card for the POPS computer is installed and press the power switch to turn on the instrument. The instrument should power up and LED 13 should begin blinking after about two minutes. You should also be able to hear the pump operating once the LED begins to blink. To shut down the instrument, press the power switch again.

Once the inlet and battery pack are installed the foam box can be closed and taped shut with electrically conductive aluminum foil tape. The system is now ready to be connected to the rest of the balloon payload, powered on, and launched.

Operation: Operation is similar to that described above for non-balloon versions, but the external power switch is used to turn the system on prior to launch. Data is stored on the SD card that is inserted into the POPS computer. The POPS-1110 comes with a 4-position XDATA cable for connecting to either an iMet RSB radiosonde or another XDATA type instrument. This cable should be connected to either the radiosonde or next instrument in the payload series.

Data transferred via the radiosonde has some configuration options, but the current standard is to provide the sample flow rate, total particle concentration, and particle count data binned over 8 or 12 size channels every second. The POPS has been implemented into NOAA's publicly available SkySonde software, which records the real-time data received from the radiosonde, POPS, and other instruments into the payload (additional hardware is required to receive data transmitted from the radiosonde; contact Handix Scientific for more information).

Outdoor Enclosure version (POPS-1120):

Setup: The POPS-1120 includes the base POPS unit inside a weatherproof enclosure, along with many accessories to power and operate the POPS in outdoor environments. The outdoor enclosure version is similar to that of the balloon version, with an included power switch that can turn on/off the POPS if being powered via battery. However, the outdoor enclosure is also able to be powered via the included 12V power supply, which connects to a dedicated weatherproof connector on the exterior of the box. The outdoor enclosure version is also default-configured to write data to the included usb drive that is mounted on the exterior of the box opposed the SD card in other configurations. However, the SD card must remain installed on the POPS for proper functionality.

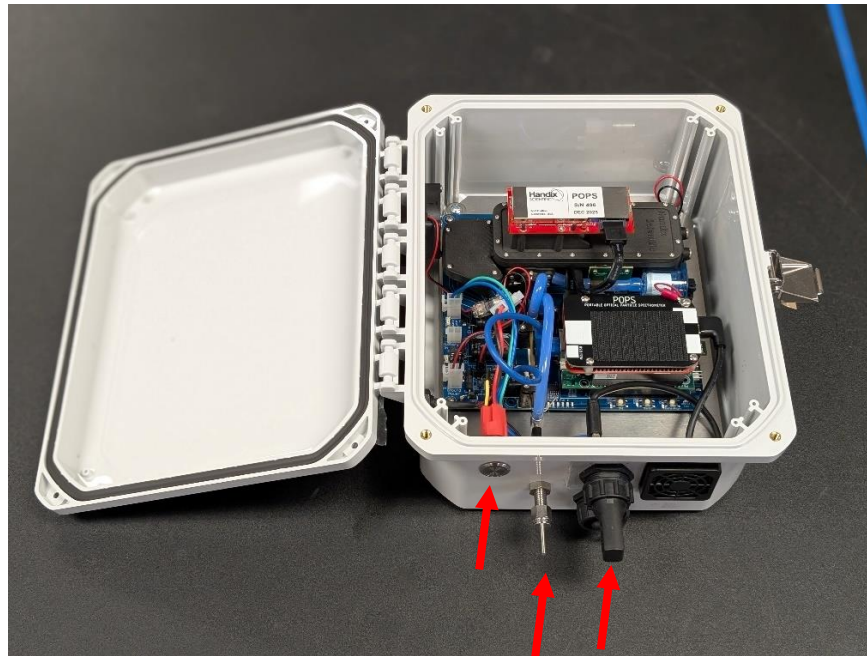


Figure 9. POPS-1120 (outdoor configuration) showing the power switch (left), sample inlet (middle) and usb drive cover (right).

Before operation, remove the inlet shipping cover that has “REMOVE ME” printed on the side. If operating the POPS-1120 on ground power, simply plug in the provided power supply to the power connector on the side of the box. When supplied ground power, the POPS will automatically turn on and begin operation, meaning the power switch is not functional and will not turn the POPS on/off when pressed. To turn off the POPS when supplied with ground power, unplug the power supply from the box. To operate on battery power, plug in the battery to the BATTERY POWER connector on the board and press the power switch to turn the instrument on. The switch LED should illuminate, indicating the POPS is receiving power. To turn the instrument off, simply press the button again. This configuration also has the ability for the POPS to be connected via ethernet for real time viewing.

When operating the POPS-1120, do not attempt to plug in ground power while the power switch is pressed and connected to the battery. This will attempt to power the POPS from both sources, leading to potential damage of the instrument and/or damage to the battery and increased risk of a battery fire. It is recommended when operating on ground power to unplug the battery from the POPS.

Viewing POPS data in real-time

The POPS Real Time Viewer software is loaded onto the USB drive that is shipped with the POPS and can be found on www.handixscientific.com/downloads.

Connecting to the POPS Beaglebone Black single-board computer

The POPS computer comes with the POPS data acquisition software pre-installed and will automatically begin operating the instrument and collecting data when the system is powered on. Users wishing

to view data in real-time will need to access the POPS Computer using an external PC. The external PC can be connected to in two ways:

Preferred Option: MiniUSB cable should be connected to the POPS Computer using the included Mini-USB cable or standard ethernet cable with RJ45. Connecting to the POPS Computer with Mini-USB allows for automatic detection within the software, making connecting convenient and easy. Connecting via Ethernet requires more involved setup, as described below. Shown in Figure 10 is the location of both the Mini-USB and Ethernet ports on the POPS. Depending on the model, some versions of the POPS only are intended to be connected via ethernet. The POPS-1120 for example only has the ethernet port connected to the exterior of the box, making ethernet connection the only viable option to connect to the POPS Computer while the lid of the POPS-1120 is closed.

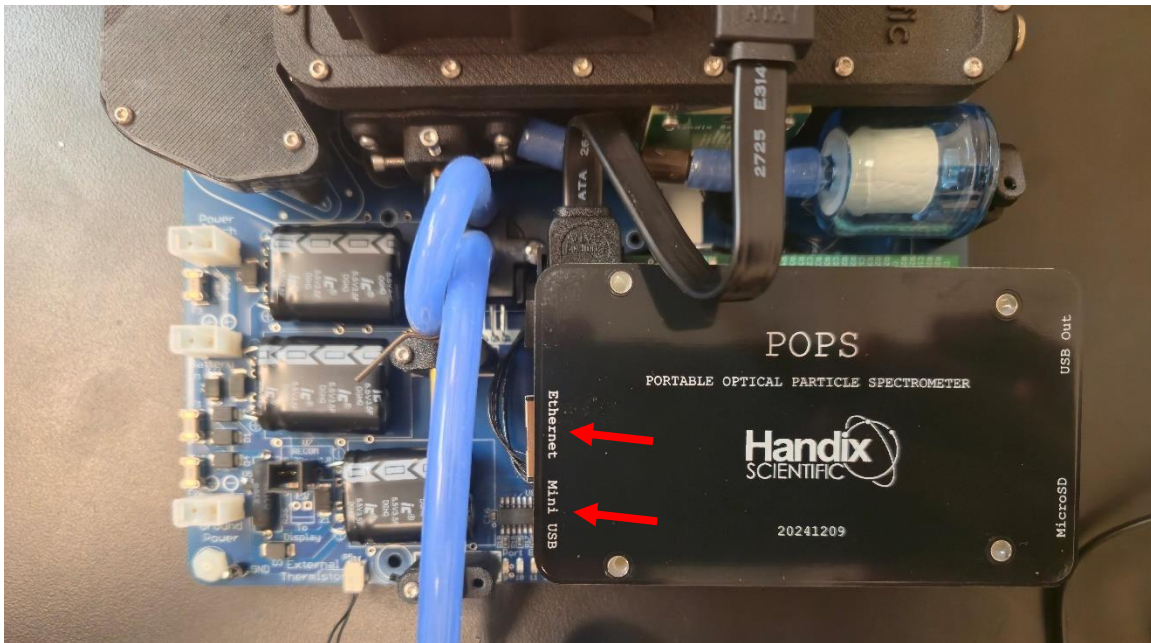
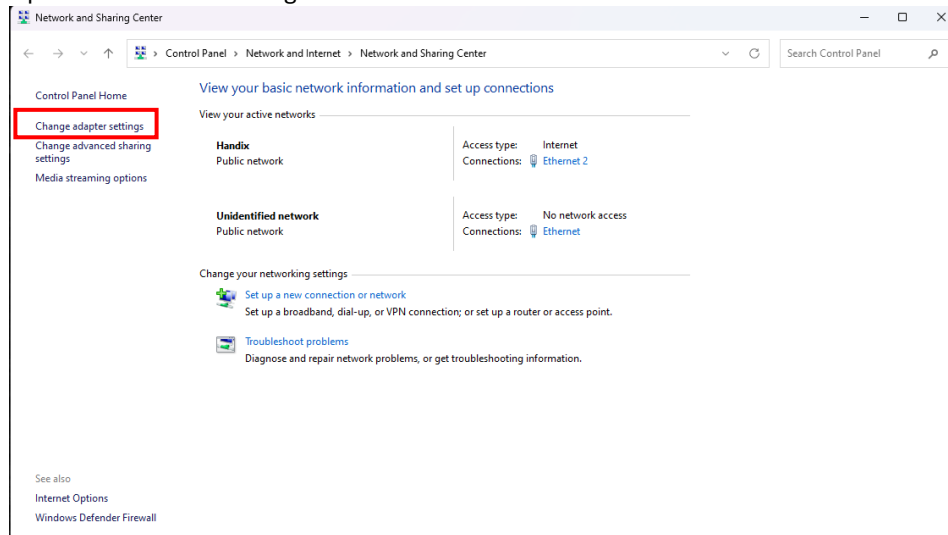


Figure 10. Locations of the Ethernet and Mini USB ports to connect POPS to external PC.

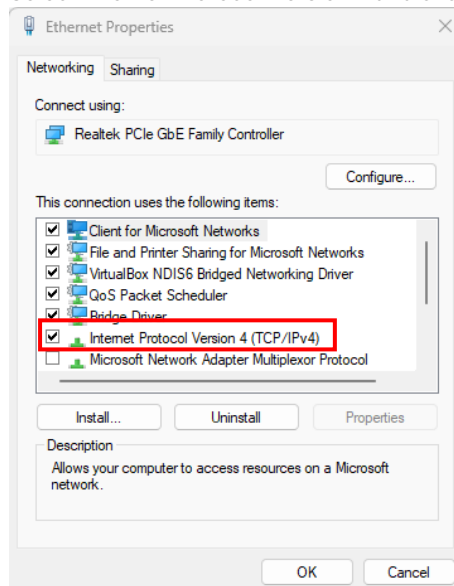
Mini usb connectusb connection steps:

When connecting to the POPS with ethernet, the POPS Computer will only accept incoming connections and provide real-time data to a specific network address, so you must edit the IP settings on the external PC as follows (for Windows 11):

1. Open Network and Sharing Center



2. Select “Change adapter settings” on the left side of the window
3. Right-click on the Local Area Connection (ethernet) associated with the POPS and select Properties. Removing and re-connecting the Ethernet cable can help identify the appropriate connection if more than one is listed.
4. Select Internet Protocol Version 4 and click Properties.



5. Select “Use the following IP address” and change the IP address to 10.11.97.100 and the subnet mask to 255.255.255.0. Select “Use the following DNS server addresses” and set the Preferred DNS server to 8.8.8.8 . Press OK to accept the changes and update the adapter settings.
6. You can test the connection to the POPS Computer by opening the command prompt in Windows and entering the command “ping 10.11.97.50” and verifying the external PC gets a reply from the POPS.


```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\middl> ping 10.11.97.50

Pinging 10.11.97.50 with 32 bytes of data:
Reply from 10.11.97.50: bytes=32 time<1ms TTL=128
Reply from 10.11.97.50: bytes=32 time<1ms TTL=128
Reply from 10.11.97.50: bytes=32 time<1ms TTL=128
Reply from 10.11.97.50: bytes=32 time<1ms TTL=128

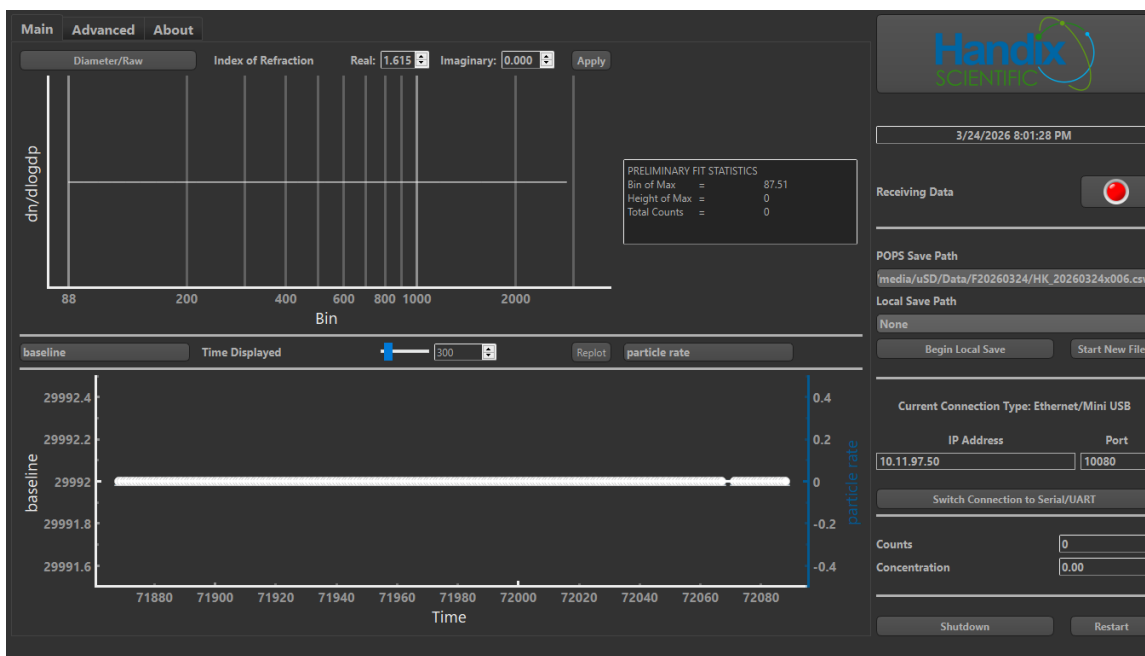
Ping statistics for 10.11.97.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\middl>

```

POPS Real Time Viewer

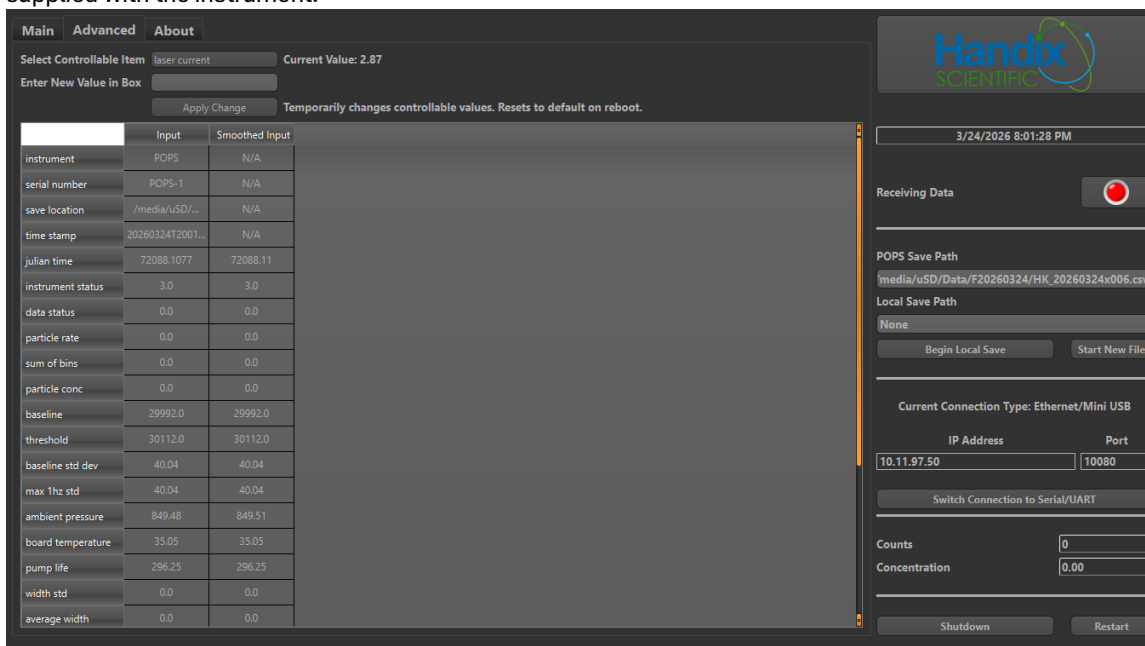
The POPS Real Time Viewer main tab is the main user interface to display information about the running instrument, including communication status, instrument timestamp (UTC), particle count rate and concentration, and the histogram of the latest one-second particle counts. The main tab also contains the ability to change the save location of data during real-time viewing using the “Local save path” and “begin local save” features.

<u>Panel Item</u>	<u>Description</u>
Instrument time	Current UTC time and date reported by POPS
IP address	IP address of the POPS computer
UDP Port	UDP port number
Serial Com Port	Serial port number on local computer connected to POPS
Start New File	Manually initiate a new file on the POPS data system
Particle Count Hist.	Display for current (1-second) particle counts, shown against either diameter or raw signal intensity depending on user selection
Left Timeseries Selec.	Data parameter to display in red against left-axis in the time series plot
Right Timeseries Selec.	Data parameter to display in blue against right-axis



POPS Advanced Control tab

The Advanced tab shows additional data parameters as well as provides access to adjust instrument configuration parameters used by the POPS **temporarily**. Parameters that can be edited appear in the “Select Controllable Item” category. Any changes made here will not be saved between power cycles on the POPS. The POPS will reset back to the values saved in the configuration file. To modify settings permanently change the values in the configuration file in the POPS configuration file on the microSD card supplied with the instrument.



Control parameters allow for custom adjustments of the POPS laser, histogram, and flow settings, and should only be adjusted by advanced users. The **laser current** parameter controls the input current to the POPS laser diode and can affect the output laser power. The **flow set** changes the target sample flow set

point (in units of cm^3 / sec). **blstart** sets the initial baseline applied when the acquisition software first loads and should always be set to a value much higher than the expected baseline. The default value is 30,000 A/D counts and **should not be modified unless instructed**. **th_mult** applies a constant multiplier to the measured standard deviation in the detector baseline signal and is used to set the trigger point for recording valid particle pulse data. Reducing the value will lead to higher data recording rates and potentially triggers on false particles or electronic and/or background light “noise”, while larger values will prevent the system from acquiring data for real particles measured at the lower range of the instrument detection window. The histogram bin parameters, **number of bins**, **logmin**, and **logmax** control the one-second particle size histograms produced by the instrument. Contact Handix Scientific before changing the **logmin** or **logmax** parameters, and use caution when setting **number of bins** to high ($n > 32$) values due to non-monotonic relationships between instrument response and particle size. The parameters **minpts** and **maxpts** set the limits for the number of digitized raw signal points allowed when classifying a raw signal pulse as a valid particle count.

POPS Data Files

The POPS automatically generates directories on the micro SD card or usb stick (model dependent) based on the file information prefix in the configuration file. By default, the instrument creates a directory for each calendar day in the “Data” directory (e.g., F20161016) containing data files and a copy of the current configuration file for that date. Files are saved depending on configuration settings. Normally it records a “housekeeping” or “HK” file containing the most commonly needed parameters and data. If desired, the system will record high time resolution raw signal data to help evaluate system performance, including laser beam profiles and baseline stability, parameterized raw peak data for individual particles such as maximum peak signal amplitude and peak width, and a log file to help diagnose any instrument problems.

Housekeeping files

Most commonly needed data are saved in comma-delimited housekeeping files, which are named “HK_YYYYMMDDxnnn.csv” where YYYY is the year, MM and DD are the month and day and nnn is a file index. Multiple files are usually collected per day if raw peak data is also being saved. The following parameters/data are saved in the HK file at 1-second time resolution:

<u>Parameter</u>	<u>Description</u>
DateTime	Date and time in seconds from midnight, 1 January 1970.
Status	1 = run
PartCt	Particle counts (# / s)
PartCon	Particle concentration (# / cm^3 ; volumetric)
BL	Current baseline of the detector (raw analog-to-digital counts)
BLTH	Current threshold for particle counting; equals $\text{BL} + \text{STD} \times \text{TH_Mult}$
STD	Current standard deviation of baseline
P	Ambient pressure measured on POPS Cape (hPa)
TofP	On-board temperature measured on POPS Cape (C)
POPS_Flow	Current sample flow rate measured by the laminar flow element and differential pressure sensor (cm^3 / s)
PumpFB	Feedback value used when controlling pump speed using PID control.

LDTemp	Temperature of laser diode control board (C)
LaserFB	Feedback value used when controlling laser power using PID control.
LD_Mon	Laser diode output power monitor
Temp	External thermistor value (not used)
BatV	Battery or input DC power voltage (V)
Laser_Current	Laser diode current (constant if not using PID control; from config)
Flow_Set	Pump voltage set point (constant if not using PID control; from config)
BL_Start	Starting point for baseline value determination
TH_Mult	Threshold multiplier for determining valid particle counts (from config)
Nbins	Number of bins for particle raw peak signal distributions (from config)
Logmin	Power of 10 for lowest particle signal bin (from config)
Logmax	Power of 10 for largest particle signal bin (from config)
Skip_save	Value of the “skip_save” parameter set in configuration file (see above)
MinPeakPts	Minimum number of peak points above threshold to be considered valid particle count (from config)
MaxPeakPts	Maximum number of peak points above threshold to be considered valid particle count (from config)
RawPts	Number of points saved for any raw data recorded
[HISTOGRAM] or [b0]	Nbins columns where Nbins is the number of histogram bins. Each column contains counts recorded in last second within a logarithmically spaced size bin. See “Determining sizing bin boundaries” in Appendix 2.

Other Files: Peak, RawBL, and RawPeak files

Peak files contain several parameters describing the raw particle peak signals recorded by the photodetector. The data can be useful if you need to re-bin to different size bins or work with > 1 Hz data. RawBL and RawPeak files contain high-speed raw signals recorded during particle and baseline periods. They are typically used for diagnosing problems with the instrument. All these files are binary files and require the correct decoding sequence. Contact Handix Scientific if you would like to work with peak, RawBL, or RawPeak data files

Data Storage Management

NEVER REMOVE THE SD CARD WHILE THE INSTRUMENT IS RUNNING!

The 32 GB microSD card (or USB drive) supplied with the POPS has sufficient storage space for up to three weeks of continuous data collection with full data writing options selected (including peakby-peak, baseline and raw data being saved), though the exact capacity will depend on particle concentrations. Reducing the save rate for peak-by-peak data or disabling raw/baseline data saving will increase storage capacity. **The POPS will no longer save newly acquired data once there is no longer space on the microSD card!** To avoid data loss, periodically move files from the microSD card to another storage device.

If the microSD card becomes full it will need to be re-formatted for the POPS to operate normally. To reformat the microSD card:

1. First copy all data, including configuration files, contents of the “Data” directory, and the pump life data file to another storage device or computer. A microSD to SD or microSD USB reader is useful for this step.
2. If using Windows, after copying data, reformat the microSD card by right-clicking on the

card in Windows Explorer and selecting “Format...” from the menu. Select the FAT32 file system and then click “Start” to format the SD card (all data will be lost at this point, so be sure data have been backed up first!).

3. If using OSX, start Disk Utility, select the microSD card, then press the “Erase” button, again, selecting the FAT32 file system.
4. After formatting the microSD card, copy the POPS_Configuration.ini and pump file (if present) back to the card, and also create a directory/folder on the card called “Data”.
5. Eject the card and insert back into the microSD card slot on the Beaglebone Black attached to the POPS. The POPS should now operate normally again.

Advanced Software Setup

The instrument configuration can also be directly edited on the micro-SD card using a card reader and basic text editor, though this is not recommended as in some cases the formatting of the file can be altered depending on the text editor used to make the configuration changes.

Accessing the file system through SSH (ethernet):

ssh [root@10.11.97.50](ssh://root@10.11.97.50)

password: B1gb0\$\$!

Accessing the file system through SSH (mini usb):

ssh [root@192.168.7.2](ssh://root@192.168.7.2)

password: B1gb0\$\$!

Alternatively, you can access data files and the configuration file directly on the micro-SD card using a card reader. **NEVER REMOVE THE SD CARD WHILE THE INSTRUMENT IS RUNNING!** Always perform a safe shut down of the system before removing the SD card. Do not change any of the files or organization of the directory structure without first consulting Handix Scientific, with the exception of the configuration file as described below. On the micro-SD card all data acquired during operation will be contained in the “Data” folder, organized and time stamped which will be described later in this manual.

Users comfortable with the linux command line can edit the file directly on the SD card while the instrument is running by connecting to the instrument with ssh, then using a text editor such as nano or vi on the POPS Computer to modify the file directly. For example, using nano the command would be “nano /media/uSD/POPS_BBB.cfg”.

Editing the configuration file

The configuration file is called “POPS_BBB.cfg” and is located on the root (also known as the “main”, “parent” or “top”) level directory on the micro-SD card. **Always keep a copy of the configuration file backed up before making any changes to the file on the micro-SD card.** The file is read by the acquisition software and used to set a number of parameters when operating the instrument. Configuration settings are described in detail in the “POPS Data Acquisition Software” section of the manual. If editing the file externally save changes and copy the file back to the SD card, overwriting the previous version.

The POPS Configuration file and default settings

<u>Parameter</u>	<u>Default Value(s)</u>	<u>Description</u>
gBaseAddr	/media/uSD/Data/F	Default directory and file name prefix for data files. Do not change.

BBB_SN	-	Serial number for the Beaglebone Black single-board computer in the instrument.
POPS_SN	-	Serial number for the instrument. Do not change
Flow	offset = X.X divisor = X.X	Coefficients used to convert the differential pressure sensor raw reading to flow rate in vccm.
gBins	Nbins = 16 logmin = 1.6 logmax = 4.817	Number of log-spaced bins for particle size distributions power of 10 for lowest bin boundary (~25 raw counts from digitizer) power of 10 for upper bin boundary (~65k raw counts from digitizer)
AI	Multiple	No user-editable values.
AO	Name = "Laser Current" set_V = 2.867 maxV = 4.096 minV = 0 Ki = 1 use_pid = false	Analog output control settings for laser current. Default is to hold laser current to constant value by applying a set voltage, which converts to approximately 70 mA and corresponds to a laser power of about 70 mW at the laser output and approximately 30 mW in the particle detection region. Contact Handix Scientific before changing.
AO	Name = "Flow_Set" set_V = 2.35 maxV = 4.096 minV = 0 Ki = 1 use_pid = false	Raw pump voltage to apply to obtain desired sample flow set point. Instruments are pre-configured to achieve a target sample flow rate of ~180 cm ³ / minute, volumetric. Increasing the set_V value will give a higher sample flow rate and reducing the set_V will give a lower sample flow rate.
SavePeakBinary	true/false	Option for saving peak-by-peak particle data in binary format. Caution, setting to "true" will create larger files and fill up the uSD card faster!
UseHistSum	true/false	Option to set real-time concentration display to be the sum of binned counts divided by sample volume (true) or the sum of all particles measured above the instrument threshold (false).
FlowStepsUse	true/false	Option to force the system to operate at higher sample flow rates at low pressure recommended for high-altitude balloon measurements. Contact Handix Scientific for more information or to change the values for the "Flow Steps" parameter.
Serial_Port	Multiple	No user-editable values.
Skip	Skip_Save = 0	Reduces number of particles saved to the particle-by-particle peak information file. Default is to save all particles, but can generate large amounts of data. Replace "0" with integer with number of particles to skip. For example, "Skip_Save=4" will save 1 of every 5 detected particles to the file and "Skip_Save=9" will save 1 of every 10.
Peak	MinPeakPts=5 MaxPeakPts=255	Number of digitized points in the raw detector signal that must be above baseline to be considered a valid particle. Do not change without first consulting Handix Scientific.
Baseline	BL_Start = 30000 TH_Mult = 3.0	BL_Start should not be changed. The value of TH_Mult determines the minimum raw signal peak height that must be measured in order for a signal to be counted as a valid particle. The signal must be greater than the average baseline plus TH_Mult standard deviations. For example, if the baseline is 1000 raw A/D counts with standard deviation of +/- 5, the default TH_Mult value (3.0) gives a signal threshold of 1015 raw A/D counts.
Raw	Multiple	Option to record raw digitized data to the uSD card. Set the "save" value to false in order to preserve uSD card space for longer duration operation.
Status	Status_Type="UAV"	Sets the operating mode of the instrument. Currently two modes are supported: UAV or iMet. UAV mode allows any number of bins (max recommended is 200) and preserves information when sent via UDP or serial to an external computer. "iMet" mode forces the system to use and transmit 8 bins, but this can be modified based on bandwidth availability.
UDP	IP port type use	IP address used for UDP data transfer. port number used for UDP data transfer S = simplified UDP data format; F = full UDP data format. The viewer is configured to read these formats automatically and display data. Contact Handix Scientific if you wish to know more details about the data streamed via UDP.

Set to false to disable UDP data transfer if you do not wish to use this feature.

Adjusting the date and time

The factory time on a POPS is UTC time. The easiest way to detect if time drift occurs is by connecting the POPS to an external PC, running the POPS Real Time Viewer, and noting the time within the Viewer and comparing it to the time listed on the external PC. If time drift happens within the POPS computer, update the clock on the POPS computer to UTC time with the following command in the command line (known programs that work with this command: MobaXterm, PuTTY, iTerm [Mac]):

When connected via ethernet:

```
ssh root@10.11.97.50 "sudo date -s @$(date -u +"%s"); sudo hwclock -w -f /dev/rtc1; sudo hwclock -w -f /dev/rtc;"
```

When connected via mini usb:

```
ssh root@192.168.7.2 "sudo date -s @$(date -u +"%s"); sudo hwclock -w -f /dev/rtc1; sudo hwclock -w -f /dev/rtc;"
```

Troubleshooting adjusting date and time: If a message pops up that end in

“Are you sure you want to continue connecting (yes/no/[fingerprint])?”

Respond ‘yes’, hit enter.

Password: B1gb0\$\$!

If you have more than one POPS, depending on your terminal configuration, you may get an error like this:

```
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@  WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!  @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ECDSA key sent by the remote host is
SHA256:Z3A17jZ1KchuWRVMhKGyD9dKwxYH/HIUPYkHh90/RKU.
Please contact your system administrator.
Add correct host key in /Users/CoolScientist/.ssh/known_hosts to get rid of th
is message.
Offending ECDSA key in /Users/CoolScientist/.ssh/known_hosts:6
ECDSA host key for 10.11.97.50 has changed and you have requested strict checking
```

In this error message, the line “Offending ECDSA key in /Users/CoolScientist/.ssh/known_hosts”: 6 indicates that the bad key is on line six of the known_hosts file (your file will likely have a different line). Delete the offending line in the known_hosts file (or the known_hosts file entirely, with discretion) and re-run the command, following the steps above.

iMet Communications Protocol

POPS-1110 units are configured by default with the ability to send important data through the XDATA port on the POPS Cape for Balloon/ radiosonde application.

If you connect the UART #1 output or the iMet/XDATA output on the POPS to a [serial terminal program](#) you will see data like this:

```
xdata=3801013B1D00012200A0005900420047004C007A0008
```

xdata= is packet header

All data are ASCII formatted MSB-first hexadecimal numbers

38 is POPS instrument ID

01 is daisy chain index

013B is total particle concentration for all size bins

1D is 10xflow_rate

00 is 100+temperature_{external thermistor}

0122... is the concentrations for each of the 8 size bins. Each concentration is represented by a 4 digit hex, so there are 32 green characters in total.

Every field after the daisy chain index is user data of any length up to the iMet's maximum bandwidth (about 80 bytes/second, aka 160 hex characters/second total for xdata instruments). Transmission is at 3.3v TTL UART levels at 9600 baud, 8 data bits, 1 stop bit, no parity, and (optional) XON/XOFF flow control.

Maintenance and Calibration

Replacing the particle filter

The particle filter (Figure 1, labeled 'sheath air input filter') should be replaced periodically, depending on use. In polluted environments, the filter should be replaced monthly. In clean environments, the filter can be replaced yearly.

1. Loosen the screw holding the filter into the filter-mounting block. You should be able to push the filter out of the holding clamp so that it can be removed.
2. Pull flexible tubing away from filter inlet. If necessary you may need to cut the tubing at the inlet to remove.
3. Insert inlet of new filter into the flexible tubing.
4. Push the filter back into the filter-mounting block and slightly tighten the screw to secure in place.

Performance checks and calibration

The POPS alignment and performance is most easily verified by providing it with monodisperse particles of a known size and distribution, such as polystyrene latex spheres. It is recommended that at minimum weekly single-size PSL checks are performed for the POPS, as the raw signal response will decrease as the collection mirror becomes dirty and the amount of light focused onto the detector decreases.

If available, a differential mobility analyzer or alternative particle classifier can be used to provide a known input distribution of particles of a known composition. Note that because the optical sizing is sensitive to the refractive index of the sampled particles, the POPS reported optical size reflects the choice of calibration material.

PMT Gain Adjustment and Verification Procedure

Applicable to POPS-1100, POPS-1110, and POPS-1120

Purpose

This procedure describes how to verify and adjust the photomultiplier tube (PMT) gain on the Portable Optical Particle Spectrometer (POPS). Proper gain adjustment ensures optimal signal-to-noise performance and accurate particle sizing across the instrument's operating size range (approximately 130 nm – 3 μ m). The PMT gain is adjusted using the trimmer potentiometer (trimpot) located on the signal processing/digitizer board mounted on top of the optical chamber.

Target Calibration Value

The PMT gain must be set such that a 495 nm polystyrene latex (PSL) calibration particle produces a measured particle size that aligns at 495 nm (± 5 nm) in the POPS Viewer interface.

Required Equipment

- 495 nm PSL calibration spheres
- Nebulizer or aerosol generator
- Optional: Diffusion dryer
- Optional: neutralizer
- Clean dilution air
- Small insulated screwdriver
- ½" heat shrink tubing approx 2mm ID
- POPS Data Viewer

Note: a fast and easy way to acquire this equipment if you do not have it is to purchase a Portable Aerosol Generation System (PAGS), Diffusion Dryer, and POPS-CAL-KIT with PSL from Handix Scientific!

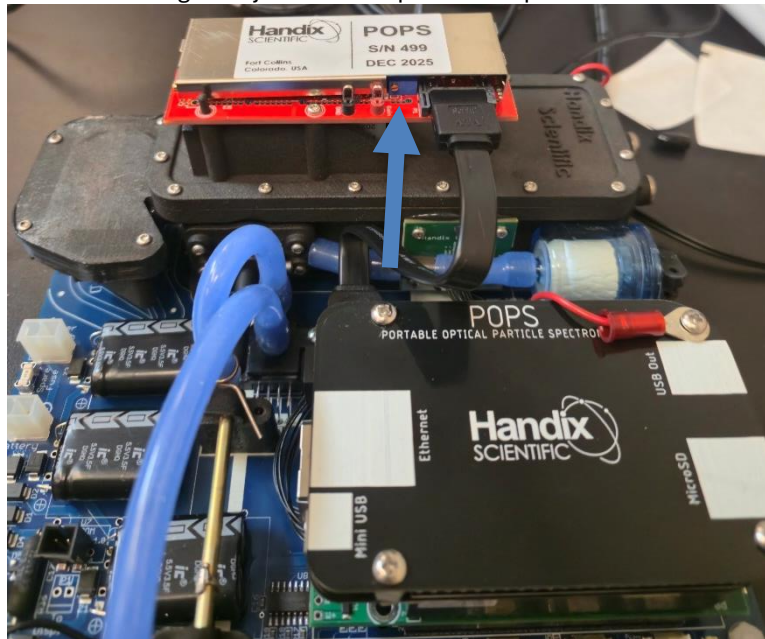
Pre-Adjustment Setup

Before beginning gain adjustment:

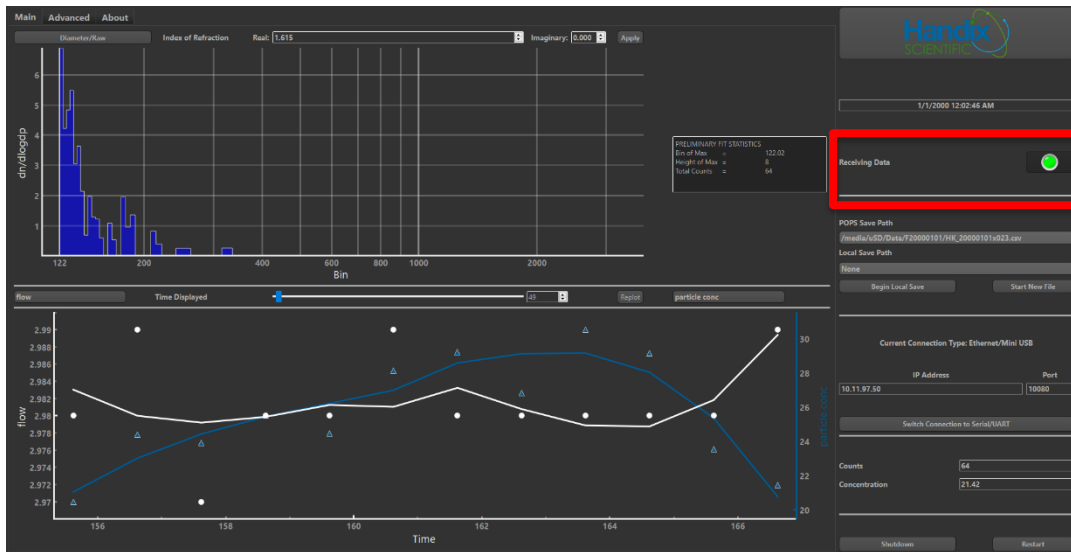
1. Install heat shrink tubing over the end of the screwdriver as shown in Figure.
 - This ensures proper seating on the trimpot.
 - Prevents accidental electrical shorting.
 - Reduces risk of slipping during fine adjustments.



2. Locate the PMT gain adjustment trimpot on the optical chamber lid



3. Connect the POPS instrument to the computer via Mini-USB or Ethernet.
4. Launch the POPS Viewer software and confirm:
 - Instrument is connected.
 - Data is actively updating.
 - No communication errors are present.



5. Verify:

- POPS inlet is unobstructed.
- Sample flow is operating at nominal value (~180 cm³/min unless otherwise configured).
- Baseline is stable.
- Baseline standard deviation is within normal operating range.

6. Allow the instrument to warm up for 5–10 minutes.

Histogram Resolution Adjustment

To improve gain alignment precision:

1. Navigate to the **Advanced** tab.
2. Under *Select Controllable Item*, select **Number of Bins**.
3. Enter **200** in the value box.
4. Select **Apply Change**.
5. Confirm the Current Value updates to 200.

Increasing the number of bins to 200 provides finer histogram resolution, allowing more precise peak alignment during gain adjustment.

Main **Advanced** **About**

Select Controllable Item: Current Value: 64.0

Enter New Value in Box:

Temporarily changes controllable values. Resets to default on reboot.

	Input	Smoothed Input
instrument	POPS	N/A
serial number	POPS-200	N/A
save location	/media/uSD/...	N/A

Generating Calibration Aerosol

1. Generate nominal 495 nm PSL aerosol using a nebulizer.
2. Ensure aerosol is properly dried prior to entering the POPS.
3. Do not pressurize the POPS inlet.
 - Use a Tee fitting with a filtered side port to avoid overpressure.
4. Maintain moderate particle concentration:
 - Recommended < 500 particles per second to avoid coincidence errors.
 - Avoid coincidence.
5. Allow concentration and flow to stabilize before adjustment.

Viewing Particle Size During Gain Adjustment

Gain adjustment is performed using the **measured particle diameter output**, not raw peak amplitude. Observe the particle size distribution in the POPS Viewer histogram.

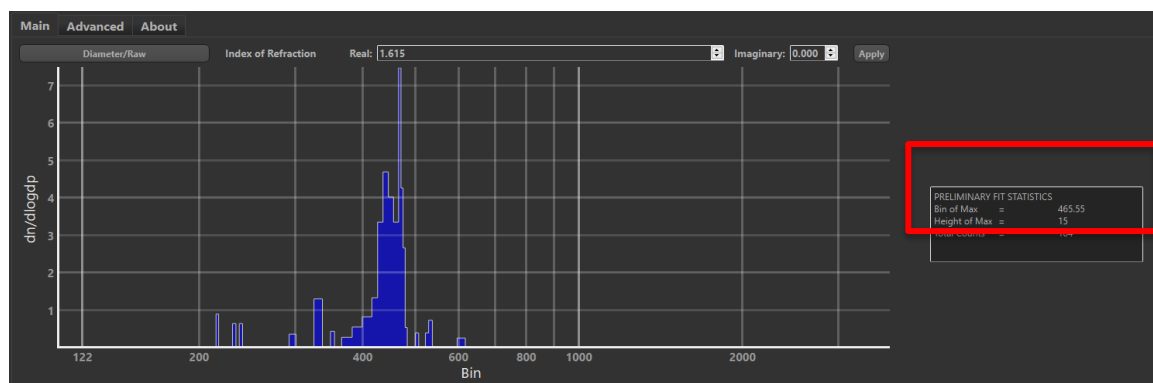
Example:

If the measured peak is centered at approximately 460 nm, the right-hand display may show:

Bin of Max = 465.55 nm

This indicates the gain must be increased.

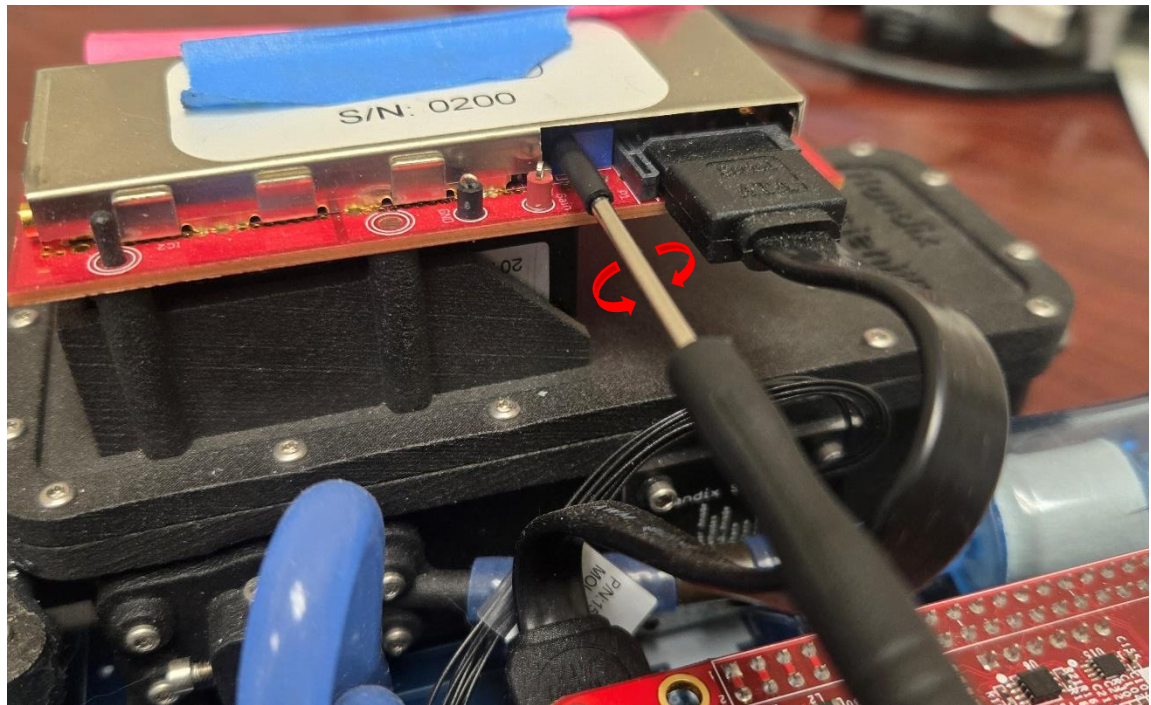
Peak position, not raw signal amplitude, is the primary adjustment metric during production calibration.



Gain Adjustment Procedure

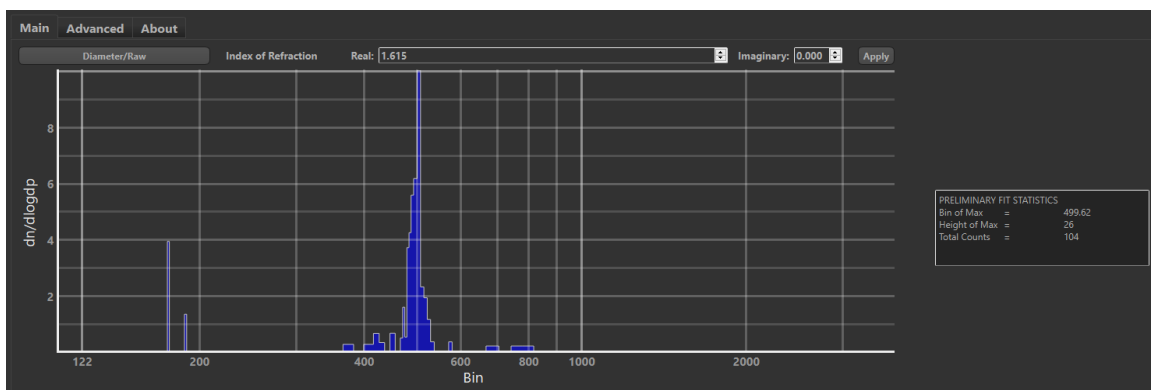
1. With stabilized 495 nm PSL aerosol flowing into the instrument, observe the peak location in the histogram.

2. Using the insulated screwdriver, adjust the PMT gain trimpot slowly.



3. The trimpot is highly sensitive:
 - Make small adjustments (~20 degrees per turn).
 - Wait for the peak to stabilize between adjustments.
4. Continue until the particle size peak aligns at:

495 nm $\pm$ 5 nm



5. Confirm:
 - Peak is stable.
 - No clipping or saturation occurs.
 - Baseline remains stable.
 - Histogram shape appears normal.
6. Optional:

- Repeat process with varying sizes of PSL within the size range of POPS to confirm accurate sizing.

Post-Adjustment Verification

After alignment:

1. Continue sampling 495 nm PSL.
2. Verify peak stability over several minutes.
3. Confirm baseline standard deviation remains within normal range.
4. Confirm histogram behavior is consistent across bins.

Important Considerations

Effect of Gain on Detection Threshold (D50)

Increasing PMT gain does not only shift particle sizing. It also affects the instrument's lower detection efficiency.

As gain is increased:

- Signal amplitude for all particles increases.
- The effective trigger threshold (baseline + threshold multiplier) corresponds to a larger equivalent optical diameter.
- The instrument D50 (diameter at 50% detection efficiency) may increase.

Excessive gain adjustment may therefore improve alignment at 500 nm while degrading performance near the lower detection limit.

For this reason, gain should be adjusted only as necessary to align the 495 nm PSL reference peak and not increased beyond what is required for proper sizing.

- Re-check gain after hardware replacement and prolonged use.

When to Perform Gain Adjustment

- Detector board replacement
- PMT replacement
- Optical system servicing
- Calibration deviation
- Directed by Handix Scientific support

When to Request Handix Support

If proper sizing cannot be achieved within the normal adjustment range of the PMT trimpot, further evaluation may be required.

Contact Handix Scientific for diagnostic support if any of the following conditions occur:

- The trimpot reaches its adjustment limit and the 495 nm PSL peak cannot be aligned.
- Excessive gain is required to reach proper sizing.

- Peak amplitude appears unstable or inconsistent.
- Baseline noise is abnormally high.
- Sizing drift persists after repeated adjustment.
- Particle sizing behavior appears nonlinear or distorted.

In some cases, inability to achieve proper sizing may indicate reduced laser output power, optical misalignment, or component degradation.

Handix Scientific can provide diagnostic guidance and determine whether further service or component replacement is necessary.

Appendix 1: Troubleshooting and FAQ

Before reaching out for help, most POPS issues can be resolved with a few quick checks. These are the most common problems users run into in the field or lab. If none of the steps below fix the issue, it's best to review the full manual and/or contact Handix Scientific for further support.

1. Check the micro-SD card
 - a. Make sure it is fully inserted. The POPS will not operate without it.
2. Verify power source
 - a. Confirm battery is connected and charged, or 12V DC power is properly plugged in
3. Wait at least 1-2 minutes after powering on
 - a. POPS takes time to boot and start data acquisition
4. Look for normal LED behavior
 - a. Ensure status LED 13 is blinking
 - b. If LED 14 is illuminated (error LED), contact Handix Scientific
5. Listen for the pump
 - a. The pump should start within ~60 seconds and reach a steady speed.
6. Check all electrical cables and connections
 - a. Power and data cables should be secure. Check battery, switch, and accessory connections to the POPS Baseplate
7. Check all flow connections
 - a. Inlet and sheath flow tubing should be unobstructed, and pump connections should be secure. If possible, measure or ensure there is airflow through the inlet of the instrument.
8. Verify connection and operation with POPS Real Time Viewer
9. Ensure required files are present on micro-SD card or USB drive
 - a. Make sure the configuration, pump file, and Data folder must remain intact.
10. Make sure the SD card or USB drive is not full
 - a. The POPS will stop saving data. Offload or clear space.

Appendix 2: Theory of Operation

Airborne particles interact with incident light depending on the particle's size, shape and composition. Light encountering an airborne particle can diffract or refract, referred to as light scattering, or be absorbed (light absorption). The amount and directional dependence of light scattering is described by Raleigh Theory for particles small relative to the wavelength of the incident light and by Lorenz-Mie Theory for spherical particles with similar sizes compared to the wavelength of the incident light. Both theories predict a strong size-dependence on the magnitude of the scattered light, meaning the light scattering signal can be converted to a physical diameter under certain assumptions. The POPS, like other optical particle counters, relies on the relationship between particle diameter and scattered light intensity to count and size individual particles.

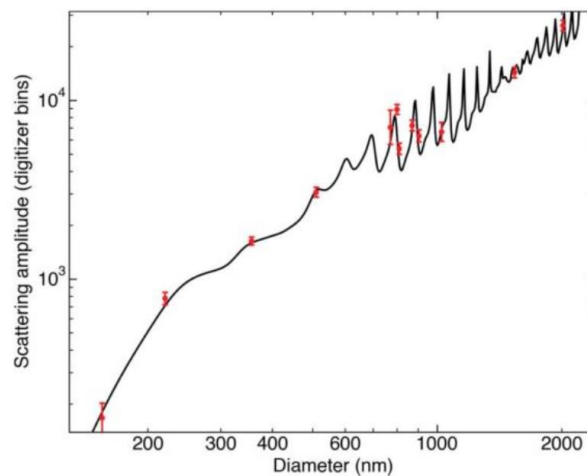


Figure 11. Scaled scattering intensities predicted by Mie theory for the POPS geometry for PSL particles ($RI = 1.615 + 0.001i$). Red points show individual PSL calibration measurements. Figure courtesy NOAA and Aerosol Science and Technology.

Figure 11 shows an example theoretical response curve predicted by Mie theory for particles with the same refractive index (RI) as polystyrene latex spheres ($1.615 + 0.001i$). The curve initially follows the diameter to the sixth power relationship predicted by Rayleigh scattering theory while the particle diameter is small compared to the POPS laser wavelength (405 nm). Oscillations in the curve become more pronounced and the size dependence becomes weaker for particle diameters near to and larger than the laser wavelength. The oscillations cause a reduced size resolution for particles at the upper range of the POPS.

Mie theory predicts a decrease in scattering intensity as the particle refractive index decreases and/or the particle becomes more strongly light absorbing (or an increase in the complex component of the refractive index). Two particles with the same size and shape but different compositions may be sized differently depending on their size and refractive index.

During operation the POPS samples particles from the ambient air through an inlet with a laminar flow element to monitor the volumetric sample flow rate. A software-controlled miniature rotary vane vacuum pump pulls air out of the optical chamber to draw flow through the inlet jet. Filtered sheath air is also pulled into the chamber by the pump, but restricted by an orifice. The pump speed is adjusted to maintain a constant volumetric flow rate through the sample inlet, as measured by the laminar flow element. Sheath

air is introduced to the sample cavity through six small jets surrounding the inlet nozzle, which helps confine the sample jet and prevent contamination of optical surfaces inside the optical chamber.

Sampled particles are focused through the center of a vertically elongated laser beam with a Gaussian beam profile. Light scattering by particles as they transit through the beam is collected by a spherical mirror located at the base of the cavity and focused onto the sensing element of a photodiode located on the opposite side of the optical chamber. After transiting the laser beam sampled particles are exhausted through a tube connected to the vacuum pump and filtered.

Appendix 3: Data Processing

Determining size bin boundaries

The POPS bins particles based on the logarithm of their peak scattering amplitudes. The bin boundaries are logarithmically spaced and determined based on the number of bins (*nbins*) and *logmin* and *logmax* parameters set in the instrument configuration file. For example, for the iMet configuration *nbins* = 8, *logmin* = 1.6 and *logmax* = 4.817. Therefore, the log delta is:

$$\log\delta = \frac{\log\max - \log\min}{nbins} = \frac{4.817 - 1.6}{8} = 0.402$$

And the bin boundaries in scattering amplitude space are:

Bin #	1	2	3	4	5	6	7	8
Lower	$10^{1.6}$	$10^{1.6+0.402}$	...	...	...	...	...	$10^{4.817-0.402}$
Upper	$10^{1.6+0.402}$	$10^{1.6+2 \times 0.402}$	...	...	...	...	...	$10^{4.817}$

Or

Bin #	1	2	3	4	5	6	7	8
Lower	40	100	...	...	...	...	...	26001
Upper	99	253	...	...	...	...	...	65614

The conversion to **optical diameter in nm** is based on Mie theory, the specific scattering geometry of the POPS and the refractive index of PSL (1.615+0.001i). For an 8-bin configuration the bin boundaries **in nm** are:

Bin #	1	2	3	4	5	6	7	8
Lower	115	135	165	210	350	575	1220	1990
Upper	135	165	210	350	575	1220	1990	3370

The default setting for the POPS is 16 bins, and for a 16-bin configuration the bin boundaries **in nm** are:

Bin#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Lower	115	125	135	150	165	185	210	250	350	475	575	855	1220	1530	1990	2585
Upper	125	135	150	165	185	210	250	350	475	575	855	1220	1530	1990	2585	3370

Contact Handix Scientific if you have questions about determining the size bin boundaries or mapping scattering amplitude to optical diameter

